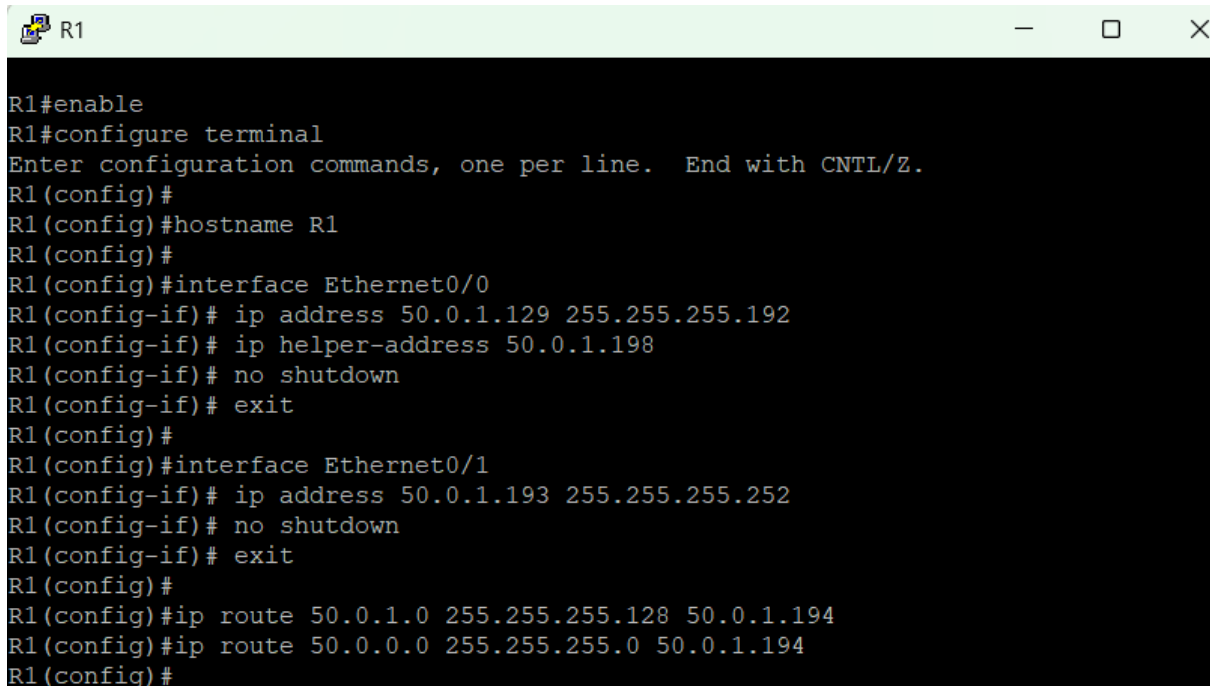
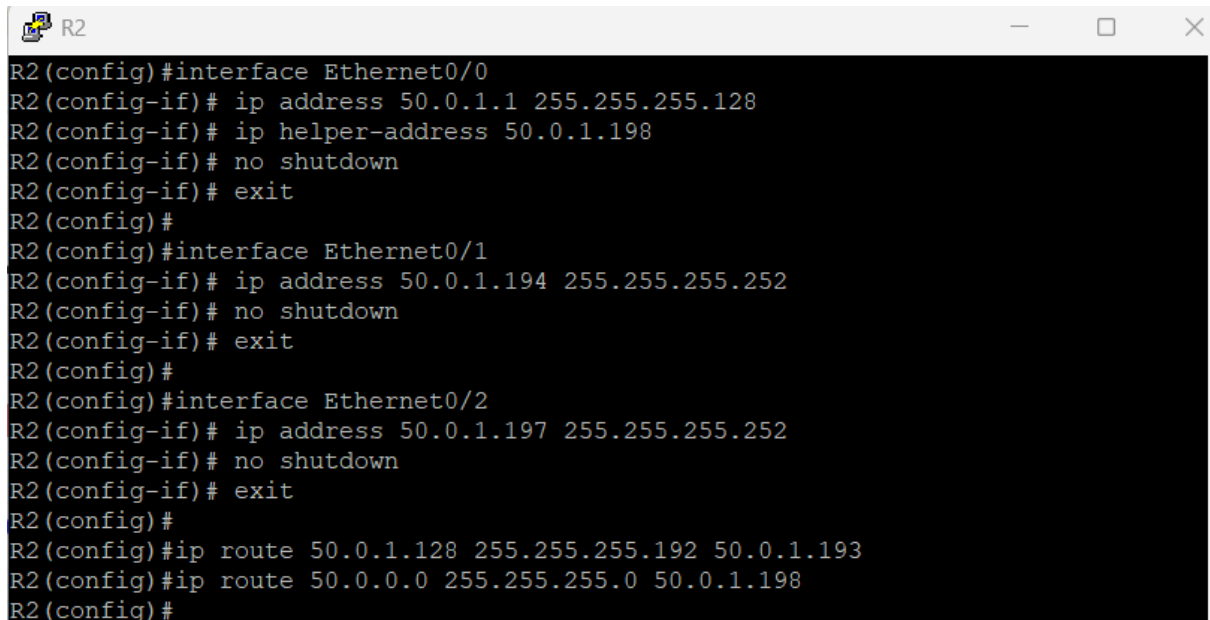


## Step 1: Assign ip for R1,R2,R3 and Enable dhcp in R3

A terminal window titled 'R1' with standard window controls. The terminal shows the configuration of R1, including enabling the terminal, setting the hostname to R1, configuring Ethernet0/0 and Ethernet0/1 interfaces with IP addresses and helper addresses, and setting static routes.

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#hostname R1
R1(config)#
R1(config)#interface Ethernet0/0
R1(config-if)# ip address 50.0.1.129 255.255.255.192
R1(config-if)# ip helper-address 50.0.1.198
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
R1(config)#interface Ethernet0/1
R1(config-if)# ip address 50.0.1.193 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
R1(config)#ip route 50.0.1.0 255.255.255.128 50.0.1.194
R1(config)#ip route 50.0.0.0 255.255.255.0 50.0.1.194
R1(config)#
```

A terminal window titled 'R2' with standard window controls. The terminal shows the configuration of R2, including configuring Ethernet0/0, Ethernet0/1, and Ethernet0/2 interfaces with IP addresses and helper addresses, and setting static routes.

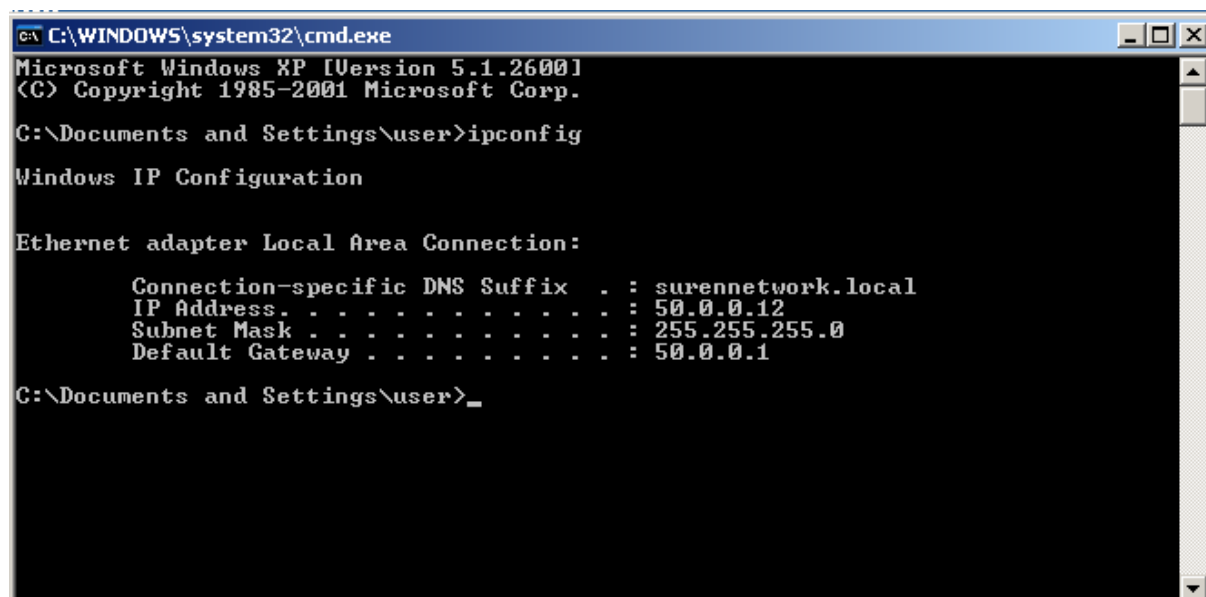
```
R2(config)#interface Ethernet0/0
R2(config-if)# ip address 50.0.1.1 255.255.255.128
R2(config-if)# ip helper-address 50.0.1.198
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#interface Ethernet0/1
R2(config-if)# ip address 50.0.1.194 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#interface Ethernet0/2
R2(config-if)# ip address 50.0.1.197 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#ip route 50.0.1.128 255.255.255.192 50.0.1.193
R2(config)#ip route 50.0.0.0 255.255.255.0 50.0.1.198
R2(config)#
```

```

R3(config)#interface Ethernet0/0
R3(config-if)# ip address 50.0.0.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#interface Ethernet0/1
R3(config-if)# ip address 50.0.1.198 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#ip route 50.0.1.0 255.255.255.128 50.0.1.197
R3(config)#ip route 50.0.1.128 255.255.255.192 50.0.1.197
R3(config)#
R3(config)#!-----
R3(config)#! Excluded Addresses
R3(config)#!-----
R3(config)#
R3(config)#ip dhcp excluded-address 50.0.0.1 50.0.0.10
R3(config)#ip dhcp excluded-address 50.0.1.1 50.0.1.10
R3(config)#ip dhcp excluded-address 50.0.1.129 50.0.1.135
R3(config)#
R3(config)#!-----
R3(config)#! SALES DHCP Pool
R3(config)#!-----
R3(config)#
R3(config)#ip dhcp pool SALES
R3(dhcp-config)# network 50.0.0.0 255.255.255.0
R3(dhcp-config)# default-router 50.0.0.1
R3(dhcp-config)# dns-server 8.8.8.8
R3(dhcp-config)# domain-name surennetwork.local
R3(dhcp-config)# lease 7
R3(dhcp-config)#
R3(dhcp-config)#!-----
R3(dhcp-config)#! PRODUCTION DHCP Pool
R3(dhcp-config)#!-----
R3(dhcp-config)#
R3(dhcp-config)#ip dhcp pool PRODUCTION
R3(dhcp-config)# network 50.0.1.0 255.255.255.128
R3(dhcp-config)# default-router 50.0.1.1
R3(dhcp-config)# dns-server 8.8.8.8

```

## Step 2: fetching ip from the pc



```

C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

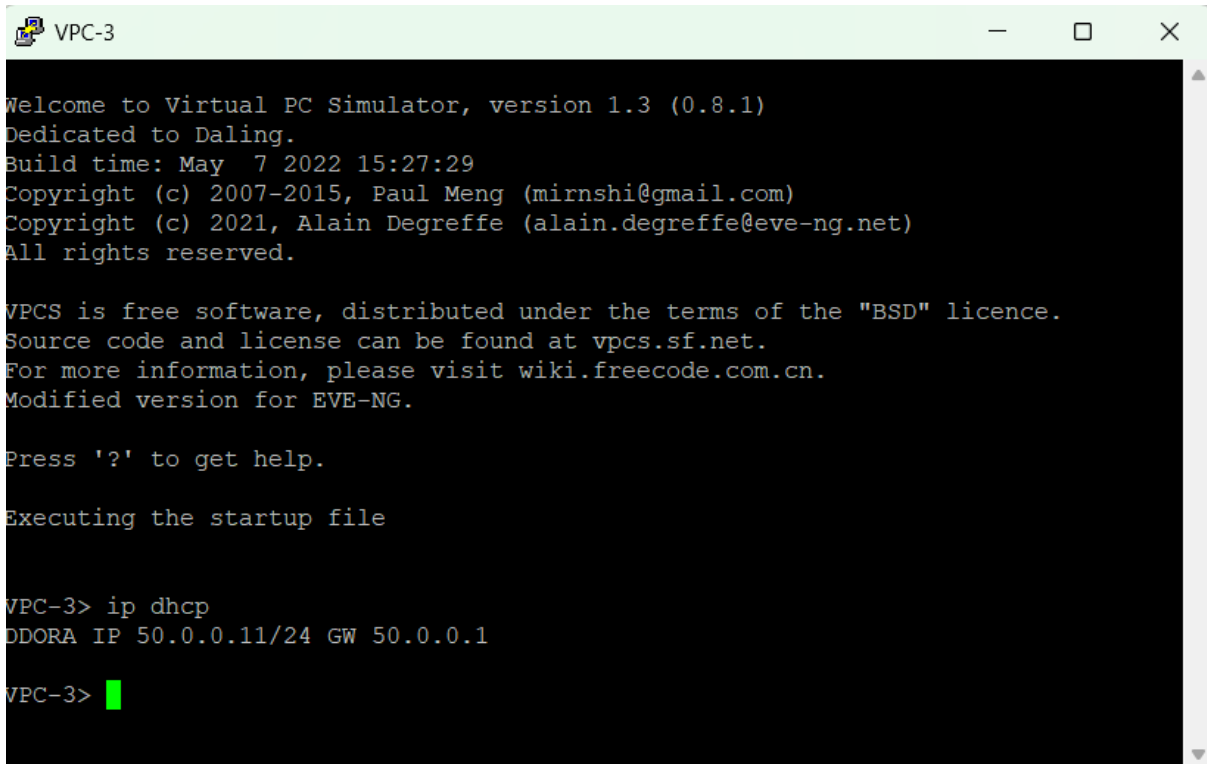
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : surennetwork.local
    IP Address. . . . .               : 50.0.0.12
    Subnet Mask . . . . .             : 255.255.255.0
    Default Gateway . . . . .         : 50.0.0.1

C:\Documents and Settings\user>_

```



The image shows a terminal window titled "VPC-3" with a light green title bar. The terminal content is as follows:

```
Welcome to Virtual PC Simulator, version 1.3 (0.8.1)
Dedicated to Daling.
Build time: May  7 2022 15:27:29
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
Copyright (c) 2021, Alain Degreffe (alain.degreffe@eve-ng.net)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC-3> ip dhcp
DDORA IP 50.0.0.11/24 GW 50.0.0.1

VPC-3> █
```